

MAT1575, Classwork1, Fall2026

4.10 Antiderivatives (p.419-429)

1. Find the derivative of the given function.

- (a) $F(x) = x^3 + x^2 + 1$ (b) $F(x) = \ln|x|$ (c) $F(x) = e^x$ (d) $F(x) = \cos(x)$ (e) $F(x) = x^3 + x^2 - 20$

(a) $F'(x) = 3x^2 + 2x$ (b) $F'(x) = \frac{1}{x}$ (c) $F'(x) = e^x$ (d) $F'(x) = -\sin(x)$ (e) $F'(x) = 3x^2 + 2x$

$\frac{dF(x)}{dx}$

2. Definition: What is an Antiderivative of a given function?

A function F is an antiderivative of the function f if, for all x in the domain of f ,

$F'(x) = f(x)$

$\frac{dF(x)}{dx} = f(x)$

Observation: The antiderivative of the function f is a function with a derivative. (The reverse of differentiation)

3. Applications: Why would we be interested in an Antiderivative of a given function?

$\frac{ds(t)}{dt}$

$v(t) = s'(t)$

$a(t) = v'(t)$

Physics(rectilinear motion): position function $s(t)$ velocity function $v(t)$ acceleration function $a(t)$

4. General Form of An Antiderivative.

Let $F(x)$ be an antiderivative of the function $f(x)$ over an interval I . Then,

$\frac{dF(x)}{dx} = f(x)$

(a) $F(x) + C$ is also an antiderivative of the function $f(x)$ over I for any arbitrary constant C .

$\frac{d}{dx}(F(x) + C)$

(b) The most general form of the antiderivative of $f(x)$ over I is $F(x) + C$.

$= \frac{d}{dx} F(x) + \frac{d}{dx} C$

5. For each of the following functions, find all antiderivatives.

(a) $f(x) = 3x^2 + 2x$

(b) $f(x) = \frac{1}{x}$

(c) $f(x) = e^x$

(d) $f(x) = \cos(x)$

(a) $F(x) = \frac{3x^{2+1}}{3} + \frac{2x^{1+1}}{2} = x^3 + x^2 + C$

(b) $F(x) = \ln|x| + C$

(c) $F(x) = e^x + C$
 $F'(x) = e^x$

(d) $F(x) = \sin(x) + C$

6. Definition: The Indefinite Integrals of a function f and the notation of integrations.

Given a function f . The indefinite integral of f is the most general antiderivative of f , denoted

$\int f(x) dx$

where the integral symbol $\int$ represents integration, $f(x)$ is called the integrand, and the symbol dx called the differential of the variable which indicates that variable of integration is x and represents an infinitesimally small change in x .

If function F is an antiderivative of f , then $\int f(x) dx = F(x) + C$

a) function
b) number
c)

7. Integration Formulas

Differentiation Formula	Indefinite Integral	Differentiation Formula	Indefinite Integral
$\frac{d}{dx}(k) = 0$	$\int k dx = k \cdot x + C$	$\frac{d}{dx}(x^n) = n \cdot x^{n-1}$	$\int x^n dx = \frac{x^{n+1}}{n+1}$ for $n \neq -1$
$\frac{d}{dx}(\ln x) = \frac{1}{x}$	$\int \frac{1}{x} dx = \ln x + C$	$\frac{d}{dx}(e^x) = e^x$	$\int e^x dx = e^x + C$
$\frac{d}{dx}(\sin(x)) = \cos(x)$	$\int \cos(x) dx = \sin(x) + C$	$\frac{d}{dx}(\cos(x)) = -\sin(x)$	$\int \sin(x) dx = -\cos(x) + C$
$\frac{d}{dx}(\tan(x)) = \sec^2(x)$	$\int \sec^2(x) dx = \tan(x) + C$	$\frac{d}{dx}(\csc(x)) = -\csc(x) \cot(x)$	$\int \csc(x) \cot(x) dx = -\csc(x) + C$
$\frac{d}{dx}(\cot(x)) = -\csc^2(x)$	$\int \csc^2(x) dx = -\cot(x) + C$	$\frac{d}{dx}(\sec(x)) = \sec(x) \tan(x)$	$\int \sec(x) \tan(x) dx = \sec(x) + C$
$\frac{d}{dx}(\sin^{-1}(x)) = \frac{1}{\sqrt{1-x^2}}$	$\int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1}(x) + C$	$\frac{d}{dx}(\tan^{-1}(x)) = \frac{1}{1+x^2}$	$\int \frac{1}{1+x^2} dx = \tan^{-1}(x) + C$
$\frac{d}{dx}(\sec^{-1} x) = \frac{1}{x\sqrt{x^2-1}}$	$\int \frac{1}{x\sqrt{x^2-1}} dx = \sec^{-1}(x) + C$		

8. Properties of Indefinite Integrals.

Let F and G be antiderivatives of f and g respectively, and let k be any real number.

(a) Sums and Differences: $\int (f(x) + g(x)) dx = \int f(x) dx + \int g(x) dx = F(x) + G(x) + C$
 $\int (f(x) - g(x)) dx = \int f(x) dx - \int g(x) dx = F(x) - G(x) + C$

(b) Constant Multiples: $\int k f(x) dx = k \int f(x) dx = k F(x) + C$

9. Evaluate each of the following indefinite integrals:

(a) $\int (5x^3 - 7x^2 + 3x + 4) dx$
 $= \int 5x^3 dx - \int 7x^2 dx + \int 3x dx + \int 4 dx$
 $= \frac{5x^4}{4} - \frac{7x^3}{3} + \frac{3x^2}{2} + 4x + C$

(c) $\int \frac{4}{1+x^2} dx = 4 \int \frac{1}{1+x^2} dx$ (or $4 \int \frac{dx}{1+x^2}$)
 $= 4 \cdot \tan^{-1}(x) + C$
 $= 4 \cdot \arctan(x) + C$

(c) $\int \frac{4}{1+x^2} dx$

(b) $\int \frac{x^2 + 4\sqrt[3]{x}}{x} dx$
 $= \int \frac{x^2}{x} + \frac{4\sqrt[3]{x}}{x} dx$
 $= \int (x + 4x^{-\frac{2}{3}}) dx = \int x dx + \int 4x^{-\frac{2}{3}} dx$
 $= \frac{x^2}{2} + \frac{4x^{\frac{1}{3}}}{\frac{1}{3}} + C$
 $= \frac{x^2}{2} + 12x^{\frac{1}{3}} + C$

(d) $\int \tan(x) \cos(x) dx$
 $\tan(x) = \frac{\sin(x)}{\cos(x)}$
 $\int \frac{\sin(x)}{\cos(x)} \cos(x) dx = \int \sin(x) dx$
 $= -\cos(x) + C$